JAN JOURNAL OF ADVANCED NURSING

ORIGINAL RESEARCH

Dietary and fluid compliance: an educational intervention for patients having haemodialysis

Shahram Baraz, Siavash Parvardeh, Eissa Mohammadi & Behrouz Broumand

Accepted for publication 10 July 2009

Correspondence to E. Mohammadi:
e-mail: mohamade@modares.ac.ir

Shahram Baraz MSc
Trainer
School of Nursing, Ahwaz Jundishapur University of Medical Sciences, Ahwaz, Iran

Siavash Parvardeh PhD
Assistant Professor
School of Medicine, Ahwaz Jundishapur University of Medical Sciences, Ahwaz, Iran

Eissa Mohammadi PhD
Associate Professor
Department of Nursing, School of Medical Sciences, Tarbiat Modares University, Tehran, Iran

Behrouz Broumand MD
Professor
Department of Nephrology, Iranian Society of Nephrology, Iran University of Medical Sciences, Tehran, Iran

BARAZ S.H., PARVARDEH S., MOHAMMADI E. & BROUMAND B. (2010) Dietary and fluid compliance: an educational intervention for patients having haemodialysis. *Journal of Advanced Nursing* **66**(1), 60–68.
doi: 10.1111/j.1365-2648.2009.05142.x

Abstract

Title. Dietary and fluid compliance: an educational intervention for patients having haemodialysis.

Aim. This paper is a report of a study conducted to determine the effect of an educational intervention on dietary and fluid compliance in patients having haemodialysis.

Background. Many of the clinical problems experienced by patients having haemodialysis are related to their failure to eat appropriate foods and restrict their fluid intake. Educational intervention in patients having haemodialysis to improve their compliance with dietary and fluid restrictions appears to be effective.

Methods. Sixty-three patients having haemodialysis in three general hospitals in Tehran, Iran, were allocated into two groups at random for oral and/or video education. They were asked to give demographic and medical data. Bimonthly average values of serum potassium, sodium, calcium, phosphate, albumin, creatinine, uric acid, and blood urea nitrogen and interdialytic weight gain were measured before and after the teaching programmes. The data were collected in 2007.

Findings. Compliance in terms of biochemical parameters and interdialytic weight gain was observed in 63·5% and 76·2% of patients in the oral and video teaching groups respectively. Statistically significant correlations were observed between demographic variables (age, educational level and occupation) and dietary and fluid compliances ($P < 0{\cdot}001$). There was no difference between the effectiveness of two educational interventions.

Conclusion. Nurses should emphasize sodium compliance in patients having haemodialysis and explain its adverse effects, such as excessive weight gain, hypertension, and peripheral oedema.

Keywords: dietary compliance, educational intervention, fluid compliance, haemodialysis, Iran, nursing

Introduction

Chronic renal failure is an irreversible and progressive renal dysfunction that may eventually lead to end-stage renal disease. Haemodialysis is one of the most important and effective treatment modalities that can help to sustain the life of such patients. In Iran, the number of patients having long-term haemodialysis has increased annually and was more than 14,000 patients in 2007 (Iranian Society for Supporting Patients having Haemodialysis, December 2007).

Many of the most common problems experienced by patients having haemodialysis are related to their non-compliance with the treatment regimens. Furthermore, inappropriate food intake and drinking excessive fluid by these patients may result in adverse effects that complicate their conditions. For example, they may experience tremors because of failure to restrict potassium levels. They may also have itching and bone pain as a result of non-compliance with phosphate restrictions (Durose *et al.* 2004). Furthermore, excessive sodium intake or drinking excessive fluid may result in excessive weight gain. Excessive sodium intake may also cause hypertension, peripheral oedema especially around the ankles, pulmonary oedema, and breathlessness (Durose *et al.* 2004, Sciarini & Dungan 1996). Moreover, high protein intake aggravates progressive renal damage (Brenner *et al.* 1982). Therefore it is apparent that compliance with dietary and fluid restrictions can not only can reduce the risk of symptoms and medical complications, but also may improve patients' quality of life (Baines & Jindal 2000).

Numerous researchers have investigated compliance among patients having haemodialysis. There is a considerable body of evidence showing that successful treatment of patients with end-stage renal disease is directly related to compliance, including dietary restrictions, medication regimens and fluid restriction (Baines & Jindal 2000, Shaw-Stuart & Stuart 2000, Ricka *et al.* 2002, Lee & Molassiotis 2002, Sharp *et al.* 2005, Denhaerynck *et al.* 2007, Barnett *et al.* 2008). However, there are many reports indicating that patients do not follow diet, fluid-intake and medication regimens (Bame *et al.* 1993, Leggat *et al.* 1998, Lee & Molassiotis 2002). In fact, patients having haemodialysis have some difficulties in accepting their condition and usually disregard the need for a therapeutic diet. This situation will remain until the adverse effects of non-compliance appear and become unbearable. If patients are acquainted with the consequences of violating dietary and fluid restrictions and if they believe those consequences to be life-threatening, they are likely to be more worried about their health. As self-care is a learned behaviour, teaching and learning are valuable procedures that can assist patients with managing their complex therapeutic regimens. However, if they are educated to understand the rationale for their medical regimens, and the causes and consequences of their choices in terms of what they eat and drink, they are able to make informed decisions about whether or not to adhere to recommendations. Therefore, it seems that lack of information is the most important factor contributing to non-compliance with therapeutic regimens, especially diet and fluid restrictions, which in turn may lead to exacerbation of the illness.

Background

The need for an adequate teaching programme about therapeutic regimens, particularly with regards to diet and fluid regimens, for patients having haemodialysis has been discussed in various nursing papers (Sciarini & Dungan 1996, Schlatter & Ferrans 1998, Ricka *et al.* 2002, Lee & Molassiotis 2002, Sharp *et al.* 2005). However, the differences between various types of educational methods and their effectiveness on dietary and fluid compliance need to be evaluated further.

Oral education consisting a one-on-one patient education session with a nephrology nurse is one of methods used for teaching patients having haemodialysis. Oral education through group teaching programme is just as effective as one-to-one teaching (Falvo 1995), and patients have stated that group education is very useful in guiding them for appropriate self-care (Ahlmén *et al.* 1993). Nevertheless, some researchers believe that patient education through oral teaching programmes is usually brief and does not provide an adequate knowledge foundation, whereas video education may have some advantages and play a supplementary role in the teaching procedures (Albert *et al.* 2007). It is apparent that using a video makes the teaching content more meaningful. By showing patients having haemodialysis what they should eat, how much fluid they can consume, and other information relevant to self-care, it would be expected that patients would retain this information better.

No study has been carried out to evaluate the effects of video education for patients with chronic medical conditions, especially end-stage renal disease. Most previous evaluations were for surgical preadmission or postsurgery education (Mahler & Kulik 1998, Mahler *et al.* 1999), self-examination (Wood *et al.* 2002), smoking cessation (May *et al.* 2003) or rape/sexually transmitted disease prevention (Acierno *et al.* 2003). As indicated above, the effect of video education on dietary and fluid compliance in patients having haemodialysis has not been studied previously.

The study

Aim

The aim of the study was to determine the effect of an educational intervention on dietary and fluid compliance in patients having haemodialysis.

Design

The current study was a randomized trial in which two teaching programmes were implemented: video education and oral education.

Participants

The data were collected in 2007 in three major general hospitals affiliated to Tehran and Iran Universities of Medical Sciences. Eligibility criteria for the study included: (a) aged older than 18 years, (b) receiving haemodialysis routinely three times a week, (c) having haemodialysis at least 6 months, (d) living in a home setting and (e) and not received any educational intervention in the past. All 155 patients having haemodialysis at the three haemodialysis centres who met the criteria were asked to participate in this study, and 63 were recruited from the three dialysis centres. They were allocated into two groups at random. The random allocation was performed using computer-generated random numbers from 0 to 99. For an equal allocation to the two groups, we took odd numbers to indicate group 1 (oral education) and even numbers to indicate group 2 (video education).

Data collection

Potential participants were approached during their dialysis sessions. The purpose of the study was explained to them and those who agreed to participate were given a consent form to sign. Those agreeing then underwent the following biochemical tests: measurement of serum potassium, sodium, calcium, phosphate, albumin, creatinine, uric acid, and BUN (blood urea nitrogen) levels and interdialytic weight gain. Average values of these parameters were obtained over the past 2 months. Patients were also asked to give demographic and medical data on age, gender, marital status, educational level, occupation and length of time on haemodialysis. The data were collected within the first 2 hours after the initiation of haemodialysis in order to ensure that patients were not suffering from any dialysis-related discomfort. After measuring biomedical parameters and collecting demographic data, patients then participated in educational sessions. Two months after educational intervention, the mean bi-monthly measurements of the above-mentioned blood biochemical parameters and interdialytic weight gain were tested again.

Educational intervention

Two educational programmes were implemented: (1) oral education through group sessions and (2) video education through showing a video to each patient. Patients in the first group (n = 32) were had the group education sessions. They were invited to attend a class on the days after their haemodialysis sessions. Two educational sessions were held in all. The duration of each session did not exceed than 30 minutes. The principal investigator (SB), a renal nurse expert, performed the teaching intervention. The group education was didactic and interactive. Participants could ask questions at the time of class. An interactive portion of teaching program was held at the end of class. In this section, patients were encouraged to offer support to each other. At the end of group sessions, each patient received a teaching booklet to take home. The teaching booklet was 'A Patient Guide to Controlling Dietary Regimen' and was based on research and supplied by Tarbiat Modarres University.

Patients in the second group took part in video education. They were individually approached during two consecutive dialysis sessions in a week. An educational film on a video disc system was shown to each patient while they were having haemodialysis. Each patient was first started on haemodialysis, and then 1–2 hours after initiation of haemodialysis and ensuring that the patient was stable and ready, they were invited to watch the 30 minute film.

The two educational interventions had similar content and covered general knowledge about end-stage renal disease and dietary management for haemodialysis, identification of restricted/non-restricted food, fluid restrictions, reasons for compliance and possible consequences of non-compliance.

Ethical considerations

The study was approved by the appropriate hospital university ethics committee. Patients were informed about the purpose of our study and those who agreed to participate signed a consent form.

Data analysis

Data were analysed with descriptive statistics using the Statistical Package for the Social Science (Version 12 for windows; SPSS, Inc.). The study was a randomized trial.

Repeated-measures analysis of variance was conducted. There was a between-subjects factor, type of teaching, with two levels: oral education and video education. There was also a within-subjects factor, time, with two levels: pretest and post-test. To find out whether or not the two teaching modalities resulted in different scores, overall means for the two teaching groups were compared. In addition, to determine whether or not the participants differed over time, overall scores were compared before and after the educational interventions. Finally, to reveal any interaction between teaching method and time, the means for each group at each time (before or after education) were analysed. As the assumptions underlying the tests were met, it was not necessary to report an epsilon correction. To compare overall mean before and after education, the ANOVA test was used, with Bonferroni correction of a P value $<0{\cdot}005$.

Correlation analyses were carried out with all the variables that showed a good association with the indicators of compliance and those that hypothetically could affect compliance.

Table 1 Participant demographics

Demographic	Mean ± SD or n (%)
Age (years)	34·85 ± 9·51
18–28	22 (34·9)
29–39	18 (28·6)
40–50	23 (36·5)
Gender	
Male	33 (52·4)
Female	30 (47·6)
Marital status	
Married	39 (61·9)
Single	24 (38·1)
Educational level	
College	33 (52·4)
Secondary or above	18 (28·6)
Primary	12 (19·0)
Occupation	
Employed	17 (27·1)
Unemployed	46 (72·9)
Length of time on haemodialysis (years)	4·6 ± 2·53

Values are mean ± SD or n (%) of 63 patients.

Results

Participant demographics

All patients with chronic end-stage renal disease and receiving maintenance haemodialysis in the dialysis units of three general hospitals in Tehran were invited to participate in the study. Sixty-three were eligible and consented to participate (Table 1), comprising 33 males and 30 females between the ages of 18–50 years (mean = 34·8 years). Twenty-three (36·5%) were aged 40 years or older and 18 (28·6%) were aged between 29 and 39 years, reflecting the ageing of the dialysis population in Tehran. Thirty-nine (61·9%) were married and 24 (38·1%) were single. Around 90% were living with their families. More than half the sample (52·4%) had at least some college education, 28·6% were educated to high school level, and 19% had completed only primary school. About one-third 17 (27·1%) were working either full-time or part-time outside home, while 46 (72·9%) were unemployed. The length of time on haemodialysis ranged from 0·5 to 8 years (mean = 4·6 years, SD = 2·53) with the majority (n = 18) on haemodialysis for 1–3 years.

Biochemical measurements after educational intervention

Overall means and standard deviations of phosphate, calcium, potassium, sodium, uric acid, creatinine, albumin and BUN for the two intervention groups are reported in Table 2.

In patients receiving oral education, creatinine, phosphate, BUN and uric acid level means decreased statistically significantly after the education. Although sodium and potassium level means decreased after oral education, these differences were not statistically significant. Also, calcium level means increased after oral education in, but not statistically significantly (Table 2).

In patients receiving video education, phosphate and uric acid level means decreased statistically significantly after education. In addition, a statistically significant increase was observed in calcium level mean in these patients. Although sodium, potassium, creatinine, and BUN level means decreased after video education, these changes were not statistically significant (Table 2).

As shown in Table 2, overall scores were compared before and after educational intervention to determine whether or not there were differences over time. The statistical analysis showed that overall mean blood levels of creatinine and interdialytic weight gain decreased and calcium increased after educational intervention, indicating changes over time.

Interdialytic weight gain changes

Mean interdialytic weight gain decreased statistically significantly in patients after both interventions (Table 2).

Table 2 Mean blood levels of biochemical parameters and overall means before and after educational intervention

Biochemical parameters	Education Groups	Before	After	*P* value
Creatinine (mg/dL)	Verbal	9·48 ± 2·75	7·36 ± 1·03	0·001
	Video	9·89 ± 2·71	8·63 ± 1·69	0·108
	Overall mean	9·68 ± 2·72	7·98 ± 1·52	0·000
Na (mEq/L)	Verbal	140·84 ± 2·56	140·42 ± 2·81	0·941
	Video	140·32 ± 3·56	139·55 ± 2·82	0·739
	Overall mean	140·58 ± 3·08	139·68 ± 2·67	0·072
K (mEq/L)	Verbal	5·46 ± 0·89	5·1 ± 0·53	0·184
	Video	5·25 ± 0·65	4·98 ± 0·67	0·420
	Overall mean	5·36 ± 0·78	5·07 ± 0·63	0·018
Ca (mg/dL)	Verbal	8·63 ± 0·8	9·04 ± 0·67	0·211
	Video	8·8 ± 1·11	9·62 ± 0·7	0·001
	Overall mean	8·71 ± 0·96	9·33 ± 0·74	0·000
Phosphate (mg/dL)	Verbal	6·16 ± 1·52	5·02 ± 0·9	0·004
	Video	6·25 ± 1·47	5·16 ± 1·21	0·007
	Overall mean	6·2 ± 1·48	5·09 ± 1·06	0·000
Uric acid (mg/dL)	Verbal	6·71 ± 1·52	5·61 ± 0·8	0·003
	Video	7·17 ± 1·29	6·32 ± 1·16	0·036
	Overall mean	6·93 ± 1·42	5·96 ± 1·05	0·000
BUN (mg/dL)	Verbal	86·39 ± 42·41	56·31 ± 9·34	0·002
	Video	93·26 ± 41·85	75·32 ± 24·93	0·139
	Overall mean	89·76 ± 41·94	65·66 ± 20·88	0·000
Albumin (g/dL)	Verbal	4·11 ± 0·27	4·18 ± 0·15	0·821
	Video	4·10 ± 0·35	4·16 ± 0·35	0·866
	Overall mean	4·11 ± 0·31	4·17 ± 0·26	0·197
IDWG (kg)	Verbal	2·97 ± 0·63	2·08 ± 0·26	0·000
	Video	3·18 ± 0·77	2·28 ± 0·41	0·000
	Overall mean	3·07 ± 0·71	2·18 ± 0·35	0·000

Values are mean ± SD (n = 32 in the verbal group and n = 31 in the video group). To compare overall mean before and after education, the ANOVA test was used with Bonferroni correction of P value < 0·005.
IDWG, interdialytic weight gain; BUN, blood urea nitrogen.

Compliance in terms of interdialytic weight gain was observed in 76·2% (n = 48) of all patients after the intervention.

Dietary and fluid compliance

Dietary and fluid compliance is shown in Table 4. Patients were classified as dietary compliant if serum creatinine, sodium, potassium, calcium, phosphate, albumin, uric acid and BUN were all within acceptable limits. Accordingly, 63·5% (n = 40) of patients were defined as dietary compliant. The remaining patients (n = 23) were non-compliant with one or more restrictions, as measured by laboratory data.

Factors associated with compliance

A statistically significant correlation was observed between age and dietary or fluid compliance (rs = −0·41, P < 0·05), with younger patients showing better compliance compared with older patients. A statistically significant correlation was also observed between educational level and dietary or fluid compliance (rs = 0·68, P < 0·01). This showed that more educated patients had better compliance compared with less educated patients. Furthermore, interdialytic weight gain was correlated with occupation (rs = 0·44, P < 0·01), as well as dietary compliance, which in turn was correlated with occupation (rs = 0·31, P < 0·05). There was no association between the demographic variables of gender, marital status, and length of time on haemodialysis with biochemical parameters.

Comparison between two educational methods

To find whether or not the two teaching modalities resulted in different scores, overall means for the two groups were compared (Table 3). However, no statistically significant difference was observed between the overall means of the biochemical parameters in the two teaching methods.

Table 3 Overall means of biochemical parameters for the two teaching groups

Biochemical parameter	Overall mean		
	Oral education	Video education	*P* value
Creatinine	8·42 ± 2·32	9·26 ± 2·33	0·095
Na	140·63 ± 2·67	139·62 ± 3·06	0·056
K	5·31 ± 0·76	5·11 ± 0·67	0·153
Ca	8·83 ± 0·76	9·21 ± 1·01	0·016
Phosphate	5·59 ± 1·36	5·70 ± 1·44	0·617
Albumin	4·15 ± 0·22	4·13 ± 0·35	0·835
Uric acid	6·16 ± 1·32	6·74 ± 1·29	0·017
BUN	71·35 ± 34·03	84·29 ± 35·34	0·058
IDWG	2·52 ± 0·65	2·73 ± 0·76	0·058

Values are mean ± SD ($n = 32$ in the verbal group, $n = 31$ in the video group). To compare overall mean before and after education, the ANOVA test was used with Bonferroni correction of *P* value $< 0{\cdot}005$. There was no significant difference between the overall means of biochemical parameters in the two teaching groups (IDWG, interdialytic weight gain; BUN, blood urea nitrogen).

Furthermore, to discover if there was any interaction between teaching and time, the means for each group at each time (before and after education) were analysed, but again no statistically significant interaction was found.

Discussion

Change in compliance

Our results demonstrated that educational intervention through either oral or video education can have effect on patients' fluid and dietary compliance. The importance of compliance is emphasized by (Barnett *et al.* 2008), who stated that 'the most well-established health care regimens are worthless if a patient chooses not to comply with the recommendations of the health care system'. Generally, there is no particular framework or standard for evaluating compliance with prescribed haemodialysis treatment. At the moment, compliance is defined by various parameters such as skipping haemodialysis sessions, shortening haemodialysis sessions, interdialytic weight gain and serum electrolytes. It has been demonstrated that many compliance measures, either laboratory or behavioural compliance indices, are associated with patient outcomes. It has been emphasized that compliance parameters should be easily measurable and verifiable, should be reproducible, clearly interpretable and should be accurate. In fact, compliance parameters should be meaningful for haemodialysis patients, and the aetiology of illness and its pathophysiological origin should be unrelated to other factors and be related to important outcomes (Kaveh & Kimmel 2001). Accordingly, dietary and fluid compliance is usually determined by measuring interdialytic weight gain and serum electrolytes (Tracy *et al.* 1987, Leggat *et al.* 1998, Lee & Molassiotis 2002).

Table 4 Dietary and fluid compliance after educational intervention

Dietary and fluid variables	Compliance (%)	
	Oral education ($n = 32$)	Video education ($n = 31$)
Creatinine	65·6 ($n = 21$)	67·7 ($n = 21$)
Sodium	46·8 ($n = 15$)	38·7 ($n = 12$)
Potassium	71·8 ($n = 23$)	70·9 ($n = 22$)
Calcium	65·6 ($n = 21$)	64·5 ($n = 20$)
Phosphate	56·2 ($n = 18$)	58·1 ($n = 18$)
Albumin	62·5 ($n = 20$)	64·5 ($n = 20$)
Uric acid	71·8 ($n = 23$)	67·7 ($n = 21$)
BUN	75·0 ($n = 24$)	70·9 ($n = 22$)
IDWG	78·1 ($n = 25$)	74·2 ($n = 23$)

IDWG, interdialytic weight gain; BUN, blood urea nitrogen.

The results obtained from the current study showed that about two-third of patients were compliant with respect to creatinine, potassium, calcium, uric acid, BUN and interdialytic weight gain.

Factors associated with compliance

The results are important in light of the fact that patients in this study were all having difficulty in controlling their sodium levels. Because salt is a major component of the most traditional Iranian foods, prescribing low-salt foods to restrict the sodium and water intake is often not acceptable to patients.

The correlation between interdialytic weight gain and occupation indicates that patients working all day or those with a part-time job, had some difficulties on controlling their weight gain between dialysis sessions. This indicates that employed patients cannot follow their nutritional regimens. Such patients often work outside and usually have time constraints, and so their diet often consists mainly of 'fast foods'. Such foods often contain salty materials such as sauce, especially white sauce and ketchup, which can result in excessive sodium intake and thirst and thus excessive fluid intake.

Our results also indicate that educational level is associated with compliance. This is consistent with other studies (Watchous *et al.* 1980, Linda & Janz 1979), and it has been suggested that knowledge may be a predictor of compliance behaviour (Peck & King 1982). However, this finding is in contrast with some other reports of no relationship between knowledge and compliance (Long *et al.* 1998, Chan & Molassiotis 1999). It has been shown that increased level of

What is already known about this topic

- Patients with end-stage renal disease must follow a rigid and complex diet.
- Dietary and fluid restrictions reduce the risk of symptoms and medical complications in patients having haemodialysis.
- Educational intervention may enhance adherence to dietary and fluid restrictions.

What this paper adds

- Educational intervention through either verbal or video education statistically significantly increases the fluid and dietary compliance of patients having haemodialysis.
- There was no difference between the effectiveness of oral vs. video education.
- Patient educational level was directly associated with compliance.

Implications for practice and/or policy

- Early vs. late outcomes of dietary and fluid non-compliance should be taken into consideration in patients having haemodialysis.
- Nurses should emphasize sodium compliance in patients having haemodialysis and explain its adverse effects, such as excessive weight gain, hypertension, and peripheral oedema.
- Establishing an empathetic relationship with patients is very important before evaluating compliance and/or any educational intervention.

patient education over time may affect the relationship between knowledge and compliance. However, knowledge may not be a sufficient factor for compliance (Schlatter & Ferrans 1998, Long *et al.* 1998, Weed-Collins & Hogan 1989). This is supported by our data showing that young and more educated patients were more compliant with the dietary and fluid regimens, and showed better compliance compared with older and less educated patients.

As shown in Table 4, the proportion of patients having dialysis who are compliant in terms of potassium restriction is higher than for those who comply with dietary restrictions to limit phosphate intake. These results, which are in consistent with other reports (Lee & Molassiotis 2002), may indicate the impact of immediate vs. delayed side effects of non-compliance with dietary restriction. Non-compliance with potassium can lead to more immediate and life-threatening consequences such as cardiac arrest and death. Thus, patients may be more aware and eager to restrict potassium intake. In contrast to potassium, non-compliance with phosphate can result in delayed adverse effects such as bone demineralization and renal bone disease, and patients will discover these consequences only after a long period of time. Thus, they may believe that it is less important and less threatening (Lee & Molassiotis 2002).

As shown in Table 2, a statistically significant increase was observed in mean calcium level after video education. It is not apparent this occurred. Although it may indicate that video education offers some advantages over oral education, at least in terms of calcium, this result not sufficient for to allow such a conclusion to be firmly drawn.

Differences between two educational methods

The results obtained from this study indicate that there was no outcome difference between the two educational methods: no statistically significant difference was observed between the overall means of biochemical parameters for the two teaching methods. Moreover, no statistically significant interaction was observed between the two major variables included teaching (oral/video) and time (before/after education). Some reports indicate that video education can improve self-care behaviours in patients with chronic medical conditions (Mahler & Kulik 1998, Mahler *et al.* 1999, Wood *et al.* 2002, May *et al.* 2003, Acierno *et al.* 2003). In these studies, however, video education was implemented alone and not compared with any other methods such as oral education. Although, it has been argued that video education offers some advantages for patients and their families (Albert *et al.* 2007), our findings suggest that there may be no difference between the two methods.

Collectively, our findings showed that an educational intervention, either through an oral or video approach, increases the fluid and dietary compliance of patients having haemodialysis.

Study limitations

In the current study, the effect of the two teaching programmes on dietary and fluid compliance in patients having haemodialysis was assessed at 2 months, rather than evaluation of the early vs. late outcomes of dietary and fluid compliance. Therefore, the long-term effect of oral vs. video education on outcomes was not studied. Also, the impact of patient counselling on their knowledge was not evaluated in the current study, and this needs to be addressed in further studies. Furthermore, it is necessary to look more closely into the relationship between knowledge and other health outcomes, as well as different aspects of oral and visual education, and the quality of haemodialysis care. Quality of life in patients having haemodialysis and its association with educational intervention is one of the most important issues that needs to be addressed in future projects.

Conclusion

Successful treatment of patients with end-stage renal disease requires the individual's compliance with a complex and critically important therapeutic regimen. Improving the knowledge of patients having haemodialysis should be integral part of treatment. Nurses should emphasize sodium compliance for these patients and explain its adverse effects, such as excessive weight gain, hypertension and peripheral oedema.

Author contributions

SB, EM and BB were responsible for the study conception and design. SB performed the data collection; supervised the study. SB, SP and EM performed the data analysis. SB and SP were responsible for the drafting of the manuscript; made critical revisions to the paper for important intellectual content; and provided statistical expertise. SB and EM obtained funding; and provided administrative, technical or material support.

Funding

This research received a grant from Tarbiat Modarres University (no grant number supplied).

Conflict of interest

No conflict of interest has been declared by the authors.

References

Acierno R., Resnick H.S., Flood A. & Holmes M. (2003) An acute post-rape intervention to prevent substance use and abuse. *Addiction and Behavior* **28**, 1701–1715.

Ahlmén J., Carlsson C. & Schönborg C. (1993) Well-informed patients with end-stage renal disease prefers peritoneal dialysis to hemodialysis. *Peritoneal Dialysis International* **13**(Suppl. 2), 196–198.

Albert N.M., Buchsbaum R. & Li J. (2007) Randomized study of the effect of video education on heart failure healthcare utilization, symptoms, and self-care behaviors. *Patient Education and Counseling* **69**, 129–139.

Baines L.S. & Jindal R.M. (2000) Non-compliance in patients receiving haemodialysis: an in-depth review. *Nephron* **85**, 1–7.

Bame S.I., Petersen N. & Wray N.P. (1993) Variation in hemodialysis patient compliance according to demographic characteristics. *Social Science Medicine* **37**, 1035–1043.

Barnett T., Li Yoong T., Pinikahana J. & Si-Yen T. (2008) Fluid compliance among patients having haemodialysis: can an educational programme make a difference? *Journal of Advanced Nursing* **61**, 300–306.

Brenner B.M., Meyer T.W. & Hostetter T.H. (1982) Dietary protein intake and the progressive nature of kidney disease: the role of hemodynamically mediated glomerular injury in the pathogenesis of progressive glomerular sclerosis in aging renal ablation, and intrinsic renal disease. *New England Journal of Medicine* **307**, 652–659.

Chan Y.M. & Molassiotis A. (1999) The relationship between diabetes knowledge and compliance among Chinese with non-insulin dependent diabetes mellitus in Hong Kong. *Journal of Advanced Nursing* **30**, 431–438.

Denhaerynck K., Manhaeve D., Dobbels F., Garzoni D., Nolte C. & De Geest S. (2007) Prevalence and consequences of nonadherence to hemodialysis regimens. *American Journal of Critical Care* **16**, 222–235.

Durose C.L., Holdsworth M., Watson V. & Przygrodzka F. (2004) Knowledge of dietary restrictions and the medical consequences of noncompliance by patients on hemodialysis are not predictive of dietary compliance. *Journal of the American Dietic Association* **104**, 35–41.

Falvo D. (1995) Educational evaluation: what are the outcomes? *Advances in Renal Replacement Therapy* **2**, 227–233.

Kaveh K. & Kimmel P.L. (2001) Compliance in hemodialysis patients: multidimensional measures in search of a gold standard. *American journal of kidney diseases* **37**, 244–266.

Lee S.H. & Molassiotis A. (2002) Dietary and fluid compliance in Chinese hemodialysis patients. *International Journal of Nursing Studies* **39**, 695–704.

Leggat J.E. Jr, Orzol S.M., Hulbert-Shearon T.E., Golper T.A., Jones C.A., Held P.J. & Port F.K. (1998) Noncompliance in hemodialysis: predictors and survival analysis. *American Journal of Kidney Disease* **32**, 139–145.

Linda B. & Janz N. (1979) Effect of a teaching program on knowledge and compliance of cardiac patients. *Nursing Research* **28**, 282–286.

Long J.M., Kee C.C., Graham M.V., Saethang T.B. & Dames F.D. (1998) Medication compliance and the older hemodialysis patient. *American Nephrology Nurses Association Journal* **25**, 43–49.

Mahler H.I. & Kulik J.A. (1998) Effects of preparatory videotapes on self-efficacy beliefs and recovery from coronary bypass surgery. *Annals of Behavioral Medicine* **20**, 39–46.

Mahler H.I., Kulik J.A. & Tarazi R.Y. (1999) Effects of a videotape information intervention at discharge on diet and exercise compliance after coronary bypass surgery. *Journal of Cardiopulmonary Rehabilitation* **19**, 170–177.

May S., West R., Hajek P., Nilsson F., Foulds J. & Meadow A. (2003) The use of videos to inform smokers about different nicotine replacement products. *Patient Education and Counseling* **51**, 143–147.

Peck C. & King N. (1982) Increasing patient compliance with prescriptions. *The Journal of the American Medical Association* **248**, 2874–2877.

Ricka R., Vanrenterghem Y. & Evers G.C.M. (2002) Adequate self-care of dialysed patients: a review of the literature. *International Journal of Nursing Studies* **39**, 329–339.

Schlatter S. & Ferrans C.E. (1998) Teaching program effects on high phosphorus levels in patients receiving hemodialysis. *American Nephrology Nurses Association Journal* **25**, 31–37.

Sciarini P. & Dungan J. (1996) A holistic protocol for management of fluid volume excess in hemodialysis patients. *American Nephrology Nurses Association Journal* **23**, 299–305.

Sharp J., Wild M.R., Gumley A.I. & Deighan C.J. (2005) A cognitive behavioral group approach to enhance adherence to hemodialysis fluid restrictions: a randomized controlled trial. *American Journal of Kidney Disease* **45**, 1046–1057.

Shaw-Stuart N.J. & Stuart A. (2000) The effect of an educational patient compliance program on serum phosphate levels in patients receiving hemodialysis. *Journal of Renal Nutrition* **10**, 80–84.

Tracy H.M., Green C. & McCleary J. (1987) Noncompliance in hemodialysis patients as measured with the MBHI. *Psychology & Health* **1**, 411–423.

Watchous .S., Thurston H. & Carter M. (1980) The nurse educator and the adult dialysis patient. *Nursing Forum* **19**, 68–84.

Weed-Collins M. & Hogan R. (1989) Knowledge and health beliefs regarding phosphate-binding medication in predicting compliance. *American Nephrology Nurses Association Journal* **16**, 278–285.

Wood R.Y., Duffy M.E., Morris S.J. & Carnes J.E. (2002) The effect of an educational intervention on promoting breast self-examination in older African American and Caucasian women. *Oncology Nursing Forum* **29**, 1081–1090.

